

DOCKHUB · REQUIREMENTS

DockHub requirements

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DOCUMENT CONTROL

Document ID	DKH-REQ-001	Version	0.8
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/dockhub		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.8	2026-10-02	Amish Chadha	Status from DKH-CAL-001 v0.6 after the approved follow-ups: R15 met with the anchors and site pad chosen; R16 USD 1,519 with them; R12 token list with anonymous prepaid cards
0.7	2026-10-02	Amish Chadha	R6 restated as about 9 % and R9 as full-rate charging up to about 30 °C ambient (DKH-DEC-001); status updated
0.6	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.5	2026-10-01	Amish Chadha	Status from DKH-CAL-001 v0.3 after the constructable design (DKH-DDR-003); R16 not met
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3. R6 relaxed to 10 % and R16 redefined to the two-bay fit (DKH-DDR-001); R2 to SwapCell interface v0.3; status from DKH-CAL-001
0.2	2026-09-25	Amish Chadha	

First measurable requirements for TRL
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0.1	2026-09-25	Amish Chadha	Initial scaffold
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DockHub requirements

These requirements are checked by calculation in DKH-CAL-001 (TRL 3). They are not yet validated with riders, hosts or a city and will be revised after co-design sessions (see DKH-PRB-001). On 2026-09-25 Amish accepted the review recommendations (DKH-DDR-001, DKH-DDR-002): R6 is relaxed from 20 % to 10 %, R16 covers the four-bay cabinet with two bays fitted, and R2 cites SwapCell interface v0.3 (DKH-DDR-001 item 12). These are decided. On 2026-10-02 Amish restated R6 as about 9 % and R9 as full-rate charging up to about 30 °C ambient (DKH-DEC-001). Every other target remains as proposed at TRL 2 and will be revised after co-design.

On paper, 11 of 16 requirements are met. R16 is over its value-engineering target; R8 and R9 are at risk; R7 and R14 cannot be verified until hardware exists.

The **design duty case** used throughout is 20 swaps a day, each returning a SwapCell pack at about 15 % state of charge, in a street location with 4.5 peak sun hours and ambient air between -10 and 45 °C. Because a swap needs an empty bay, a station with n bays holds $n - 1$ packs.

Table 1. Requirements. Status is from DKH-CAL-001.

ID	Requirement	Target	Verification	Status at TRL 3 (DKH-CAL-001)
R1	Fast swap	60 s or less from tapping the reader to closing the second door, for a rider familiar with the station	Task analysis; later timed trials with riders	Met (31 s)
R2	SwapCell compatibility	Every bay builds to SwapCell interface v0.3: envelope, guide faces, blind-mate receptacle with the 10 k Ω INTERLOCK coding resistor, class D catch, CAN heartbeat and message set; no change to the interface	Interface review with the SwapCell project	Met by design review; one CAN channel per bay
R3	Charge speed	20 to 80 % in 1.5 h or less; full charge in 2.5 h or less, per bay, at 25 °C	Charge profile calculation	Met (1.2 h and 2.3 h)
R4	Throughput	20 or more swaps a day from 4 bays at the design duty case	Day model with a demand profile	Met by the four-bay fit (20 of 20 served, longest wait 4 min; ceiling 36 a day). The two-bay first prototype serves 9 of 20
R5	Grid supply	One single-phase branch circuit (120 V 15 A or 230 V 10 A); peak draw 1.5 kW or less	Load calculation	Met (1.25 kW; 11.0 A at 120 V) with chargers of power factor 0.9 or better
R6	Solar contribution	Canopy solar supplies about 9 % of daily charging energy at the design duty case, with normal routing (restated on 2026-10-02, DKH-DEC-001; relaxed from 20 % to 10 % by DKH-DDR-001)	Day model	Met on paper, thin margin (9.3 %; 11.4 % if flat packs are routed to bay 1 in daylight, which turns one rider in 20 away, so normal routing is kept; 16.1 % if the solar output is later switched to the flattest pack, a TRL 4 trial)
R7	Fire detection and response	Detect a pack or bay over 60 °C, or smoke, within 10 s; open that bay's charge contactor within 1 s of detection; sound a local alarm and send a remote alert; vent gas to the rear roof louver, away from the user side	Latency budget; later bench test with a heat source	Not verifiable at TRL 3 (electronic latency 3.1 s, contactor 0.15 s)
R8	Fire containment	A thermal runaway in one bay does not spread to a neighboring bay or pack for 30 min or more, and flame does not leave the cabinet front	Later propagation test by an accredited lab	At risk ; not verifiable at TRL 3
R9	Climate	Enclosure IP54 or better; operate (store and swap) at -10 to 45 °C ambient; full-rate charging up to about 30 °C ambient, with daytime pauses accepted above it and no charging start for cold-soaked packs below about -1 °C; never charge a pack outside the pack's reported charge temperature window (restated on 2026-10-02, DKH-DEC-001)	Thermal calculation; later chamber test	At risk : uninterrupted charging needs ambient below about 29 to 39 °C (pack charge limit 45 °C), so the warmest bay may pause just under 30 °C; cold-soaked packs cannot start below about -1 °C, as now stated
R10	Security	1.5 mm or thicker steel; fail-secure bay locks; cabinet anchored to its pad with	Design review	Met by design review

ID	Requirement	Target	Verification	Status at TRL 3 (DKH-CAL-001)
		tamper-resistant fixings; a pack is released only against a valid token and a returned pack		
R11	Accessibility and use	All operable parts (reader, buttons, bay handles) 0.38 to 1.22 m (15 to 48 in) above the ground; status shown by light and symbol as well as text; usable with gloves and at night	Model check; later user trials	Met (0.73 to 1.10 m)
R12	Privacy and data	No cameras or microphones; logs hold pack ID, bay, time, energy and state of health, with the rider as an opaque token; operates offline for 24 h or more and syncs later	Design review	Met by design review (token list of 200 kB, including anonymous prepaid cards with a pack deposit)
R13	Sidewalk footprint	Cabinet footprint 1.0 x 0.6 m or less; canopy underside 2.1 m or more above the sidewalk	Model check	Met (1.0 x 0.5 m with a flush plinth; lowest canopy point over the sidewalk 2.17 m)
R14	Maintainability	Any charger, lock, bay cradle or controller replaced in 15 min or less by one technician with hand tools, through the service door	Design review; later timed trial	Not verifiable at TRL 3
R15	Wind and anchoring	Stays upright and intact at a 30 m/s gust with the canopy fitted	Wind load calculation	Met on paper: four M12 stainless wedge anchors (3.8 kN design tension each) into a 1,400 x 1,100 x 300 mm cast pad of about 1,109 kg; equilibrium factor 1.12 at a 30 m/s gust. The anchor rating is confirmed from the maker's data when bought
R16	Cost	Parts 1,200 USD or less for the four-bay cabinet with two bays fitted (first prototype), SwapCell packs excluded (redefined, DKH-DDR-001)	Priced BOM (bom/ bom.csv)	Over the value-engineering target by \$319 (\$1,519 against \$1,200: \$1,214 for the buildable cabinet, DKH-DDR-003, plus \$305 for the anchors and site pad chosen on 2026-10-02; \$1,743 with all four bays fitted)

■ Assumptions

- SwapCell pack (interface v0.3): 46.8 V nominal, about 468 Wh, 2.85 kg, charge on a certified 54.6 V 5 A charger in 2.3 h to full and 1.2 h for 20 to 80 %, charge window 0 to 45 °C (SWC-PRC-001 v0.3, SWC-CAL-001).
- Charger efficiency about 90 %, cell charge efficiency about 95 %, cabinet auxiliary load about 30 W on average and 40 W peak.
- 400 W panel, 4.5 peak sun hours, 0.8 derate, 96 % MPPT, feeding bay 1 only.
- Packs are released at 80 % state of charge or more, fullest first; riders arrive with a lunch and a dinner peak and leave after waiting 15 min (DKH-CAL-001 Table 1).

- Delivery riders cover roughly 60 to 100 km a day at about 10 Wh/km. This is an assumption to check with riders; it implies 1 to 2 swaps per rider per shift.
- Accessibility reach range from the US ADA forward reach limits (15 to 48 in); local rules may differ.